

Trainings ▾

General Information

⚠ CAUTION ⚠

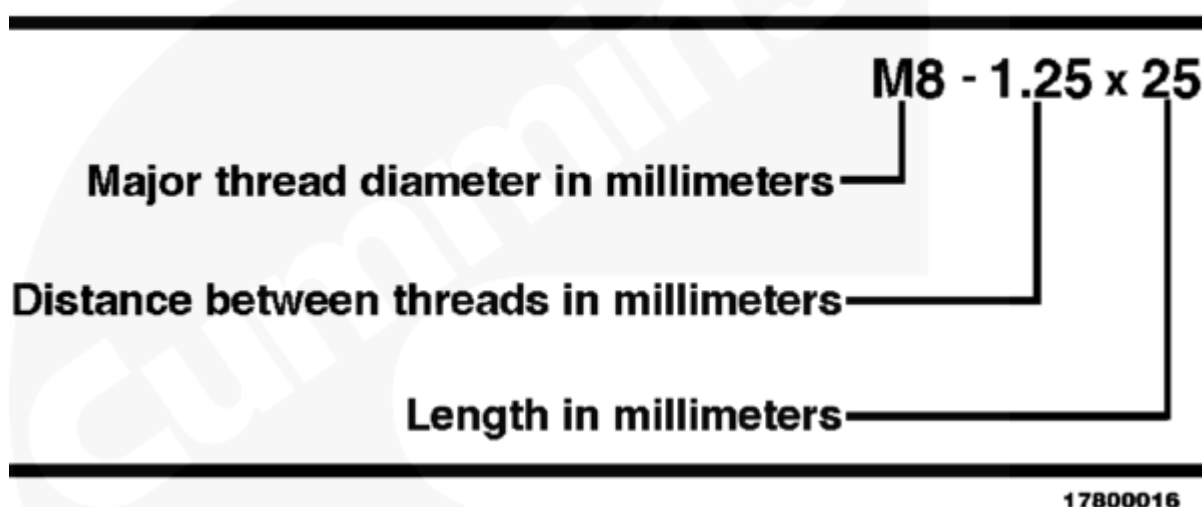
When replacing capscrews, always use a capscrew of the same measurement and strength as the capscrew being replaced. Using the wrong capscrews can result in engine damage.

Metric capscrews and nuts are identified by the grade number stamped on the head of the capscrew or on the surface of the nuts. U.S. Customary capscrews are identified by radial lines stamped on the head of the capscrew.

The following examples indicate how capscrews are identified:

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Metric - M8-1.25 X 25



- **Always** use the torque values listed in the following tables when specific torque values are **not** available.
- Do **not** use the torque values in place of those specified in other sections of this manual.
- The torque values in the table are based on the use of lubricated threads.

U.S. Customary [5/16 X 18 X 1-1/2]

Major thread diameter in inches $\frac{5}{16}$

Number of threads per inch 18

Length in inches 1-1/2

17800017

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- The torque values in the table are based on the use of lubricated threads.

Capscrew Markings and Torque Values - Metric

Commercial Steel Class					
8.8		10.9		12.9	
Capscrew Head Markings					
					

Cap scre w Gra de	8.8				10.9				12.9			
Mat erial	Cast Iron		Aluminum		Cast Iron		Aluminum		Cast Iron		Aluminum	
Cap scre w Bod y Dia met er [mm]	Torq ue [N•m]	Torq ue [ft-lb]	Torq ue [N•m]	Torq ue [ft-lb]	Torq ue [N•m]	Torq ue [ft-lb]	Torq ue [N•m]	Torq ue [ft-lb]	Torq ue [N•m]	Torq ue [ft-lb]	Torq ue [N•m]	Torq ue [ft-lb]
6	9	5	7	4	13	10	7	4	14	9	7	4
7	14	9	11	7	18	14	11	7	23	18	11	7
8	23	17	18	14	33	25	18	14	40	29	18	14
10	45	33	30	25	65	50	30	25	70	50	30	25
12	80	60	55	40	115	85	55	40	125	95	55	40
14	125	90	90	65	180	133	90	65	195	145	90	65
16	195	140	140	100	280	200	140	100	290	210	140	100
18	280	200	180	135	390	285	180	135	400	290	180	135
20	400	290	—	—	550	400	—	—	—	—	—	—

Capscrew Markings and Torque Values - U.S. Customary

SAE Grade Number	5	8
Capscrew Head Markings		
These are all SAE Grade 5 (3 line)		
		
Capscrew Torque - Grade 5 Capscrew		Capscrew Torque - Grade 8 Capscrew

Capscrew Grade	5				8			
Material	Cast Iron		Aluminum		Cast Iron		Aluminum	
Capscrew Body Diameter [in] and T.P.I.	Torque [N•m]	Torque [ft-lb]	Torque [N•m]	Torque [ft-lb]	Torque [N•m]	Torque [ft-lb]	Torque [N•m]	Torque [ft-lb]
1/4 - 20	9	7	8	6	15	11	8	6
1/4 - 28	12	9	9	7	18	13	9	7
5/16 - 18	20	15	16	12	30	22	16	12
5/16 - 24	23	17	19	14	33	24	19	14
3/8 - 16	40	30	25	20	55	40	25	20
3/8 - 24	40	30	35	25	60	45	35	25
7/16 - 14	60	45	45	35	90	65	45	35
7/16 - 20	65	50	55	40	95	70	55	40
1/2 - 13	95	70	75	55	130	95	75	55
1/2 - 20	100	75	80	60	150	110	80	60
9/16 - 12	135	100	110	80	190	140	110	80
9/16 - 18	150	110	115	85	210	155	115	85
5/8 - 11	180	135	150	110	255	190	150	110
5/8 - 18	210	155	160	120	290	215	160	120
3/4 - 10	325	240	255	190	460	340	255	190

3/4 - 16	365	270	285	210	515	380	285	210
7/8 - 9	490	360	380	280	745	550	380	280
7/8 - 14	530	390	420	310	825	610	420	310
1 - 8	720	530	570	420	1100	820	570	420
1 - 14	800	590	650	480	1200	890	650	480